Labsheeet - 10

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_absolute_error, mean_squared_error, r2_
```

```
df = pd.read_csv("/content/Housing.csv")

print(df.head())
print(df.shape)
print(df.info())
```

```
      price  area  bedrooms  bathrooms  stories mainroad guestroom basement  \
0  13300000  7420         4          2        3      yes        no       no
1  12250000  8960         4          4        4      yes        no       no
2  12250000  9960         3          2        2      yes        no      yes
3  12215000  7500         4          2        2      yes        no      yes
4  11410000  7420         4          1        2      yes       yes      yes

  hotwaterheating airconditioning  parking prefarea furnishingstatus
0              no             yes        2      yes        furnished
1              no             yes        3       no        furnished
2              no              no        2      yes   semi-furnished
3              no             yes        3      yes        furnished
4              no             yes        2       no        furnished
(545, 13)
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 545 entries, 0 to 544
Data columns (total 13 columns):
 #   Column            Non-Null Count  Dtype
---  ------            --------------  -----
 0   price             545 non-null    int64
 1   area              545 non-null    int64
 2   bedrooms          545 non-null    int64
 3   bathrooms         545 non-null    int64
 4   stories           545 non-null    int64
 5   mainroad          545 non-null    object
 6   guestroom         545 non-null    object
 7   basement          545 non-null    object
 8   hotwaterheating   545 non-null    object
 9   airconditioning   545 non-null    object
 10  parking           545 non-null    int64
 11  prefarea          545 non-null    object
 12  furnishingstatus  545 non-null    object
dtypes: int64(6), object(7)
memory usage: 55.5+ KB
None
```

```
print(df.isnull().sum())
```

```
price             0
area              0
bedrooms          0
```

```
bathrooms           0
stories             0
mainroad            0
guestroom           0
basement            0
hotwaterheating     0
airconditioning     0
parking             0
prefarea            0
furnishingstatus    0
dtype: int64
```

```
df = df.dropna()

print(df.isnull().sum())
```

```
price               0
area                0
bedrooms            0
bathrooms           0
stories             0
mainroad            0
guestroom           0
basement            0
hotwaterheating     0
airconditioning     0
parking             0
prefarea            0
furnishingstatus    0
dtype: int64
```

```
print("Duplicates:", df.duplicated().sum())

df = df.drop_duplicates()
```

```
Duplicates: 0
```

```
print(df.describe())
```

```
              price          area    bedrooms   bathrooms     stories  \
count  5.450000e+02    545.000000  545.000000  545.000000  545.000000
mean   4.766729e+06   5150.541284    2.965138    1.286239    1.805505
std    1.870440e+06   2170.141023    0.738064    0.502470    0.867492
min    1.750000e+06   1650.000000    1.000000    1.000000    1.000000
25%    3.430000e+06   3600.000000    2.000000    1.000000    1.000000
50%    4.340000e+06   4600.000000    3.000000    1.000000    2.000000
75%    5.740000e+06   6360.000000    3.000000    2.000000    2.000000
max    1.330000e+07  16200.000000    6.000000    4.000000    4.000000

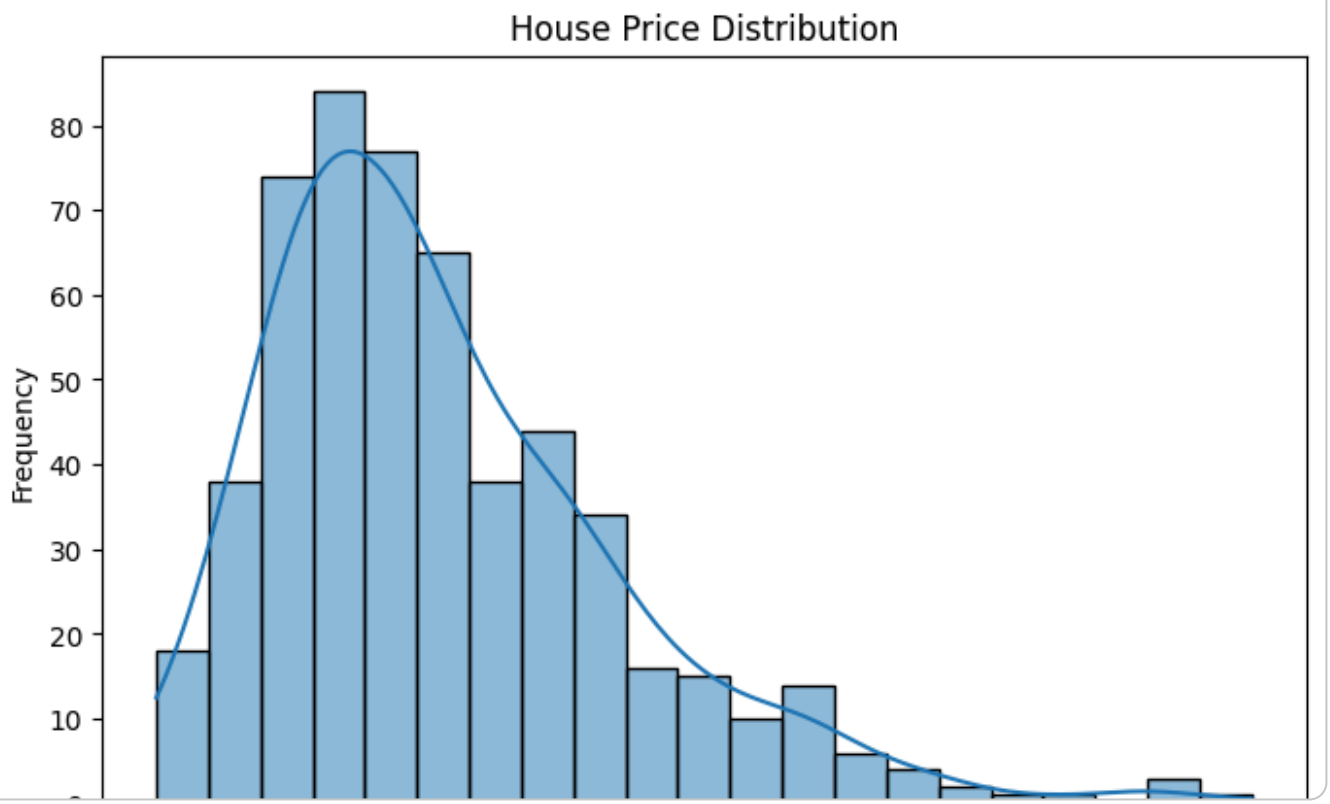
          parking
count  545.000000
mean     0.693578
std      0.861586
min      0.000000
25%      0.000000
50%      0.000000
75%      1.000000
max      3.000000
```

```
plt.figure(figsize=(8,5))

sns.histplot(df['price'], kde=True)

plt.title("House Price Distribution")
plt.xlabel("Price")
plt.ylabel("Frequency")

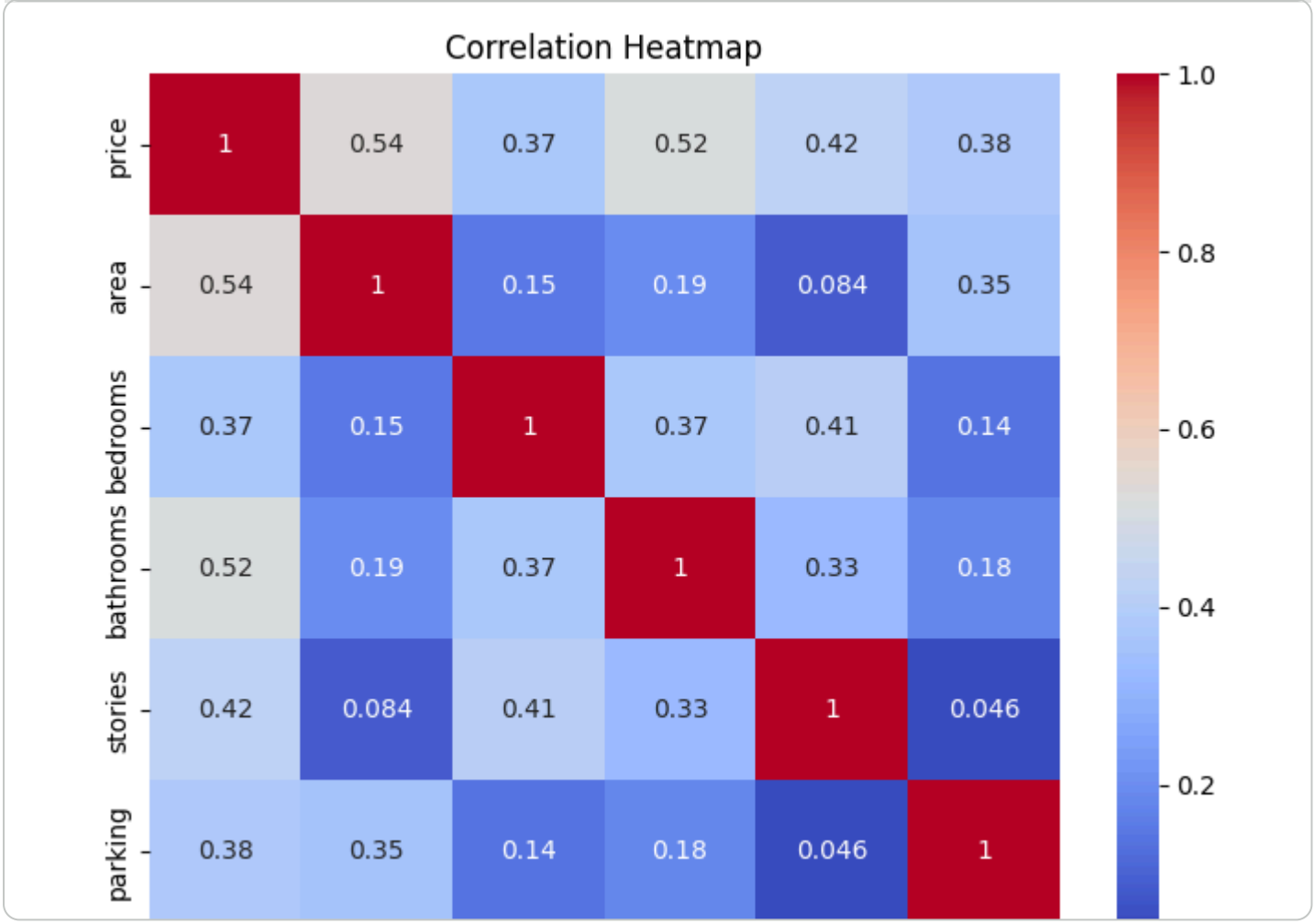
plt.show()
```




```
plt.figure(figsize=(8,6))

sns.heatmap(
    df.select_dtypes(include=np.number).corr(),
    annot=True,
    cmap="coolwarm"
)

plt.title("Correlation Heatmap")
plt.show()
```




```
X = df[['area', 'bedrooms', 'bathrooms']]
y = df['price']
```

```
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
```

```
model = LinearRegression()

model.fit(X_train, y_train)
```

▾ LinearRegression

```
LinearRegression()
```

```
y_pred = model.predict(X_test)

print(y_pred)
```

```
[6383168.05981192 6230250.35140428 3597885.28090091 4289730.8386382
 3930445.60575177 4883933.33918115 5505773.16533075 6068793.48103629
 3321512.02483442 3495157.72744705 8731338.12527607 3561265.8244721
 3487335.97847431 3633344.35548029 3933900.2714526  6661080.95290716
 2893133.47793136 4635197.40872131 4583377.42320885 4274999.75826381
 4296640.17003986 4920207.32903988 3462807.85199841 3875170.95453847
 5412497.19140831 7425564.68389184 3269692.03932195 5021117.35611339
 7122556.71331971 3238600.04801447 5366583.9261965  3640253.68688195
 6057517.06636272 4847659.34932243 4572198.51082701 5573863.86824637
 4167435.67282878 4021179.33154444 3764621.65211187 5307730.62714319
 5319221.21748587 3456243.98716683 6202613.02579763 4013357.58257171
```

```
 4534099.68582614 4235271.10270425 6057517.06636272 4525560.52386579
 4983206.22693419 3238600.04801447 6567080.25723531 3238600.04801447
 4886475.58731091 4116997.55359665 4180093.95378268 3567705.7071645
 5971150.42384195 3391517.75642211 5150855.01571622 3514973.30408096
 4723103.68827085 4722288.7729915  4427005.04910038 3915714.52537737
 4414098.80386812 3861352.29173514 5904599.35795509 3598797.69847198
 5824514.82737829 4531557.43769638 4969297.37060092 4759565.37395123
 5104119.52646327 7612116.63173673 3129432.61186821 5816602.88487562
 3816441.63762434 3865621.87271532 4635197.40872131 4393370.80966313
 6542082.68205015 3971901.59416174 5884783.78132117 4866660.010677
 3149691.15736394 7466018.06120081 3529704.38445536 3754257.65500938
 6908277.24633825 7840034.37446164 3940907.10514598 5343403.87739169
 4084993.1447181  3740438.99220605 9091535.77573358 4200821.94798766
 4935974.80912452 5988423.7523461  4462464.12367976 6624994.6588701
 3681709.67529193 5573863.86824637 3588433.70136949 6547076.98477974
 4995395.05917882 5316678.96935611 6417714.71682023 6057517.06636272
 6092063.72337104]
```

```
y_pred = model.predict(X_test)

print(y_pred)
```

```
[6383168.05981192 6230250.35140428 3597885.28090091 4289730.8386382
 3930445.60575177 4883933.33918115 5505773.16533075 6068793.48103629
 3321512.02483442 3495157.72744705 8731338.12527607 3561265.8244721
 3487335.97847431 3633344.35548029 3933900.2714526  6661080.95290716
 2893133.47793136 4635197.40872131 4583377.42320885 4274999.75826381
 4296640.17003986 4920207.32903988 3462807.85199841 3875170.95453847
 5412497.19140831 7425564.68389184 3269692.03932195 5021117.35611339
 7122556.71331971 3238600.04801447 5366583.9261965  3640253.68688195
 6057517.06636272 4847659.34932243 4572198.51082701 5573863.86824637
 4167435.67282878 4021179.33154444 3764621.65211187 5307730.62714319
 5319221.21748587 3456243.98716683 6202613.02579763 4013357.58257171
 4534099.68582614 4235271.10270425 6057517.06636272 4525560.52386579
 4983206.22693419 3238600.04801447 6567080.25723531 3238600.04801447
 4886475.58731091 4116997.55359665 4180093.95378268 3567705.7071645
 5971150.42384195 3391517.75642211 5150855.01571622 3514973.30408096
 4723103.68827085 4722288.7729915  4427005.04910038 3915714.52537737
 4414098.80386812 3861352.29173514 5904599.35795509 3598797.69847198
 5824514.82737829 4531557.43769638 4969297.37060092 4759565.37395123
 5104119.52646327 7612116.63173673 3129432.61186821 5816602.88487562
 3816441.63762434 3865621.87271532 4635197.40872131 4393370.80966313
 6542082.68205015 3971901.59416174 5884783.78132117 4866660.010677
 3149691.15736394 7466018.06120081 3529704.38445536 3754257.65500938
 6908277.24633825 7840034.37446164 3940907.10514598 5343403.87739169
 4084993.1447181  3740438.99220605 9091535.77573358 4200821.94798766
 4935974.80912452 5988423.7523461  4462464.12367976 6624994.6588701
 3681709.67529193 5573863.86824637 3588433.70136949 6547076.98477974
 4995395.05917882 5316678.96935611 6417714.71682023 6057517.06636272
 6092063.72337104]
```

```
mae = mean_absolute_error(y_test, y_pred)

rmse = np.sqrt(mean_squared_error(y_test, y_pred))

r2 = r2_score(y_test, y_pred)

print("MAE:", mae)
print("RMSE:", rmse)
print("R2 Score:", r2)
```

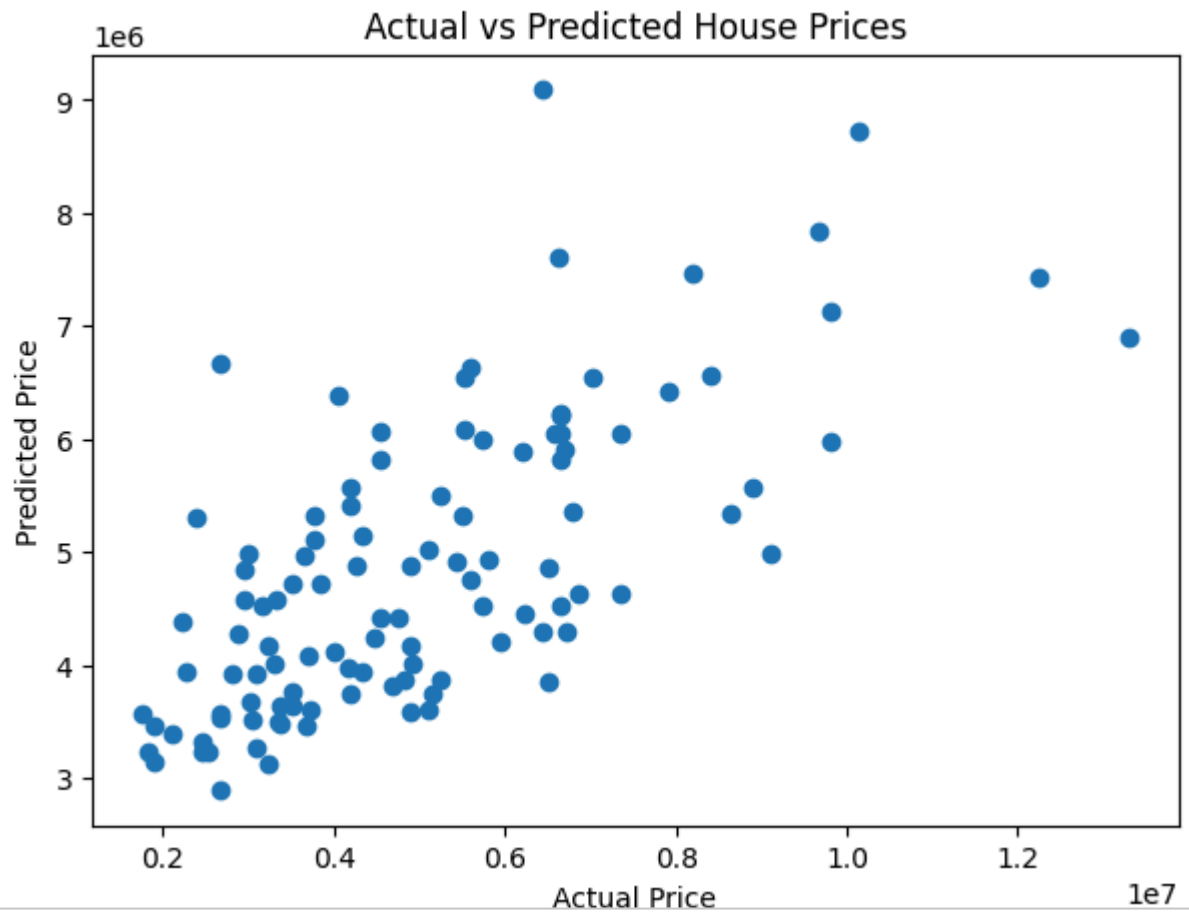
```
MAE: 1265275.6699454375
RMSE: 1658324.6001036866
R2 Score: 0.4559299118872445
```

```
plt.figure(figsize=(7,5))

plt.scatter(y_test, y_pred)

plt.xlabel("Actual Price")
plt.ylabel("Predicted Price")
plt.title("Actual vs Predicted House Prices")

plt.show()
```




```
print("""
Business Insights:
1. House price is strongly influenced by property area.
2. Number of bedrooms and bathrooms also affects price.
3. The model can help estimate prices of new properties.
4. Businesses can use these predictions for better pricing decisions.
""")
```

```
Business Insights:
1. House price is strongly influenced by property area.
2. Number of bedrooms and bathrooms also affects price.
3. The model can help estimate prices of new properties.
4. Businesses can use these predictions for better pricing decisions.
```

1. What are the major stages involved in an end-to-end data analytics project? Data collection, preprocessing, EDA, statistical analysis, visualization, machine learning, model

evaluation, and business recommendations.

2. Why is data preprocessing considered the foundation of successful analytics and machine learning? It cleans the data by handling missing values, duplicates, outliers, and categorical variables, making the data suitable for analysis and modeling.
3. How does Exploratory Data Analysis (EDA) contribute to model development? EDA helps understand data patterns, relationships, and unusual values using statistics and visualizations.
4. What factors should be considered while selecting a machine learning algorithm for a business problem? Consider the type of problem, dataset, features, required accuracy, and model performance.
5. Why is model evaluation essential before deploying a predictive model? It checks the model's performance and accuracy on data before using it for real-world decisions.
6. Explain how Business Intelligence tools complement machine learning in data analytics projects. BI tools such as Tableau and Power BI present ML results through dashboards and visualizations, making them easier to understand.
7. What challenges are commonly encountered while working with real-world datasets? Common challenges are missing values, duplicates, outliers, inconsistent data, and categorical variables.
8. How can analytical insights be converted into actionable business recommendations? By identifying important trends and using them to suggest specific actions that improve business decisions and performance.
9. Why is effective presentation and data storytelling important in analytics projects? It makes complex analytical results easier to understand and helps stakeholders make better decisions.
10. Suggest future improvements or advanced techniques that could enhance the accuracy and effectiveness of the developed analytics solution. Future improvements include better feature selection, hyperparameter tuning, cross-validation, advanced ML models, and interactive dashboards.

```
...
```

```
Ellipsis
```

```
Start coding or generate with AI.
```